Yang Zhou

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Employment

xAI, Member of Technical Staff, Kernel Team *Oct 2025 -*

Maintaining the xAI GPU communication stacks for pretraining, post-training, and inference.
Highlights: 400G/800G RDMA for GB200/GB300, fast failure recovery, low-latency allreduce, heterogeneous public cloud support (e.g., AWS EFA RDMA)

University of California, Davis, Assistant Professor of Computer Science *July 2025 -*

University of California, Berkeley, Postdoctoral researcher in Sky Computing Lab *July 2024 - now*

Project: UCCL: Building Efficient GPU Communication Systems
Supervisor: Ion Stoica

Google SRG and NetInfra, Research Intern *June 2021 - May 2023*

VMware Research, Research Intern *July 2020 - September 2020*

Meta/Facebook, Research Collaborator *November 2019 - May 2020*

SenseTime, Software Engineering Intern *March 2018 - May 2018*

Education

Harvard University, Cambridge, MA, USA

Ph.D. in Computer Science *June 2024*
M.S. in Computer Science *November 2021*
Thesis: Network-Application Co-design for Efficient Datacenters
Advisors: Minlan Yu and James Mickens

Peking University, Beijing, China

B.S. in Computer Science *July 2018*
Thesis: Towards Faster and More Accurate Data Stream Processing
Advisor: Tong Yang

Open Source Software

- UCCL, ultra and unified GPU communication library, 1300+ stars
  https://github.com/uccl-project/uccl
- LEANN, the smallest vector index in the world for RAG, 10,000+ stars
  https://github.com/yichuan-w/LEANN
- NEO, an LLM inference engine built to save the GPU memory by CPU offloading
  https://github.com/NEO-MLSys25/NEO
- eTran, implementing reliable network transports in the Linux kernel with eBPF
  https://github.com/eTran-NSDI25/eTran
- DINT, running distributed transactions in the Linux kernel with eBPF
  https://github.com/DINT-NSDI24/DINT
- Electrode, running Paxos consensus in the Linux kernel with eBPF
  https://github.com/Electrode-NSDI23/Electrode

Professional Service

Program Committee:

- ACM APSys, IEEE ICNP, USENIX NSDI, ACM EuroSys, ACM ASPLOS 2026
- USENIX OSDI 2025
- ACM SIGCOMM Workshop on eBPF and Kernel Extensions 2024, 2025
- ACM SIGCOMM Workshop on Networks for AI Computing 2025, 2026, 2027
- ACM SIGCOMM Poster/Demo 2023, 2024, 2025
- IEEE INFOCOM: Workshop on Networking Algorithms 2020

Co-Chair:

- ACM SIGCOMM Artifact Evaluation 2024

Publications

Preprints

[1] Shuang Ma, ChonLam Lao, Zhiying Xu, Zhuang Wang, Ziming Mao, Delong Meng, Zhen Jia, Jun Wu, Ion Stoica, Yida Wang, **Yang Zhou**
UCCL-Zip: Lossless Compression Supercharged GPU Communication [link]
Arxiv April 2026

[2] Expert-Choice Routing Enables Adaptive Computation in Diffusion Language Models
Shuibai Zhang, Caspian Zhuang, Chihan Cui, Zhihan Yang, Fred Zhangzhi Peng, Yanxin Zhang, Haoyue Bai, Zack Jia, **Yang Zhou**, Guanhua Chen, Ming Liu [link]
Arxiv April 2026

[3] Xingyu Xiang, Raj Joshi, Yuhan Liu, Jiayi Yao, Chenxingyu Zhao, Junchen Jiang, **Yang Zhou**, Eddie Kohler, Minlan Yu.
ShadowServe: Interference-Free KV Cache Fetching for Distributed Prefix Caching. [link]
Arxiv Sep 2025

[4] Jiarong Xing, Yifan Qiao, Simon Mo, Xingqi Cui, Gur-Eyal Sela, **Yang Zhou**, Joseph Gonzalez, Ion Stoica.
Towards Efficient and Practical GPU Multitasking in the Era of LLM. [link]
Arxiv Aug 2025

[5] Shiyi Cao, Yichuan Wang, Ziming Mao, Pin-Lun Hsu, Liangsheng Yin, Tian Xia, Dacheng Li, Shu Liu, Yineng Zhang, **Yang Zhou**, Ying Sheng, Joseph Gonzalez, Ion Stoica.
Locality-Aware Fair Scheduling in LLM Serving. [link]
Arxiv Jan 2025

[6] Yifan Qiao, Shu Anzai, Shan Yu, Haoran Ma, Shuo Yang, Yang Wang, Miryung Kim, Yongji Wu, **Yang Zhou**, Jiarong Xing, Joseph Gonzalez, Ion Stoica, Harry Xu.
ConServe: Harvesting GPUs for Low-Latency and High-Throughput Large Language Model Serving. [link]
Arxiv Oct 2024

[7] Shuo Yang, Ying Sheng, Yilong Zhao, Joseph Gonzalez, **Yang Zhou**, Ion Stoica, Lianmin Zheng.
Post-Training Sparse Attention with Double Sparsity. [link]
Arxiv Aug 2024

Conference

[8] **Yang Zhou**, Zhongjie Chen, Ziming Mao, ChonLam Lao, Shuo Yang, Pravein Govindan Kannan, Jiaqi Gao, Yilong Zhao, Yongji Wu, Kaichao You, Fengyuan Ren, Zhiying Xu, Costin Raiciu, Ion Stoica.
UCCL-Tran: An Extensible Software Transport Layer for Machine Learning Workloads. [link]
OSDI 2026

[9] Ziming Mao, Yihan Zhang, Chihan Cui, Zhen Huang, Kaichao You, Zhongjie Chen, Zhiying Xu, Zhenyu Gu, Scott Shenker, Costin Raiciu, **Yang Zhou**, Ion Stoica.
UCCL-EP: Portable Expert-Parallel Communication. [link]
OSDI 2026

[10] Yilong Zhao, Xiaonan Nie, Kan Zhu, Shuang Ma, Zhichao Lai, Hongxiang Hao, **Yang Zhou**, Baris Kasikci, Ion Stoica
Unleashing Scalable Context Parallelism for Foundation Models Pre-Training via FCP
MLSys 2026.

[11] Yichuan Wang, Shu Liu, Zhifei Li, Yongji Wu, Ziming Mao, Yilong Zhao, Xiao Yan, Zhiying Xu, **Yang Zhou**, Ion Stoica, Sewon Min, Matei Zaharia, Joseph E. Gonzalez.
LEANN: a Low-Storage Vector Index. [link]
MLSys 2026.

[12] Yilong Zhao, Shuo Yang, Kan Zhu, Lianmin Zheng, Baris Kasikci, Yifan Qiao, **Yang Zhou**, Jiarong Xing, Ion Stoica.
BlendServe: Optimizing Offline Inference for Auto-regressive Large Models with Resource-aware Batching. [link]
ASPLOS 2026

[13] Yifan Zhang, Jianchang Su, Zixu Shen, **Yang Zhou**, Wei Zhang.
Toward Interference-Aware Scheduling for Serverless Functions via eBPF and Meta-Learning. [Link]
PACMI 2025. Workshop on Practical Adoption Challenges of ML for Systems.

[14] Xiangfeng Zhu, **Yang Zhou**, Yuyao Wang, Xiangyu Gao, Arvind Krishnamurthy, Sam Kumar, Ratul Mahajan, Danyang Zhuo.
Rethinking RPC Communication for Microservices-based Applications. [Link]
HotOS 2025.

[15] Xuanlin Jiang, **Yang Zhou**, Shiyi Cao, Ion Stoica, Minlan Yu.
NEO: Saving GPU Memory Crisis with CPU Offloading for Online LLM Inference. [link]
MLSys 2025.

[16] Zhongjie Chen, Qingkai Meng, ChonLam Lao, Yifan Liu, Fengyuan Ren, Minlan Yu, **Yang Zhou**.
eTran: Extensible Kernel Transport with eBPF. [link]
USENIX NSDI 2025.

[17] **Yang Zhou**, Mark Wilkening, James Mickens, and Minlan Yu.
SmartNIC Security Isolation in the Cloud with S-NIC. [link]
ACM EuroSys 2024.

[18] **Yang Zhou**, Xingyu Xiang, Matthew Kiley, Sowmya Dharanipragada, and Minlan Yu.
DINT: Fast In-Kernel Distributed Transactions with eBPF. [link]
USENIX NSDI 2024.

[19] **Yang Zhou**, Zezhou Wang, Sowmya Dharanipragada, and Minlan Yu.
Electrode: Accelerating Distributed Protocols with eBPF. [link]
USENIX NSDI 2023.

[20] **Yang Zhou**, Hassan Wassel, Sihang Liu, Jiaqi Gao, James Mickens, Minlan Yu, Chris Kennelly, Paul Turner, David Culler, Hank Levy, and Amin Vahdat.
Carbink: Fault-Tolerant Far Memory. [link]
USENIX OSDI 2022.

[21] **Yang Zhou**, Ying Zhang, Minlan Yu, Guangyu Wang, Dexter Cao, Eric Sung, and Starsky Wong.
Evolvable Network Telemetry at Facebook. [link]
USENIX NSDI 2022.

[22] **Yang Zhou**, Tong Yang, Jie Jiang, Bin Cui, Minlan Yu, Xiaoming Li, and Steve Uhlig.
Cold Filter: A Meta-Framework for Faster and More Accurate Stream. Processing [link]
ACM SIGMOD 2018.

[23] Tong Yang, Jie Jiang, Peng Liu, Qun Huang, Junzhi Gong, **Yang Zhou**, Rui Miao, Xiaoming Li, and Steve Uhlig.
Elastic Sketch: Adaptive and Fast Network-Wide Measurements. [link]
ACM SIGCOMM 2018.

[24] Omid Alipourfard, Masoud Moshref, **Yang Zhou**, Tong Yang, and Minlan Yu.
A Comparison of Performance and Accuracy of Measurement Algorithms in Software. [link]
ACM Symposium on SDN Research (SOSR) 2018.

[25] Xiangyang Gou, Chenxingyu Zhao, Tong Yang, Lei Zou, **Yang Zhou**, Yibo Yan, Xiaoming Li, and Bin Cui.
Single Hash: Use One Hash Function to Build Faster Hash Based Data Structures. [link]
IEEE International Conference on Big Data and Smart Computing (BigComp) 2018.

[26] Tong Yang, **Yang Zhou**, Hao Jin, Shigang Chen, and Xiaoming Li.
Pyramid Sketch: A Sketch Framework for Frequency Estimation of Data Streams. [link]
VLDB 2017.

[27] **Yang Zhou**, Peng Liu, Hao Jin, Tong Yang, Shoujiang Dang, and Xiaoming Li.
One Memory Access Sketch: A More Accurate and Faster Sketch for Per-Flow Measurement. [link]
IEEE Global Communications Conference (Globecom) 2017.

[28] Junzhi Gong, Tong Yang, **Yang Zhou**, Dongsheng Yang, Shigang Chen, Bin Cui, and Xiaoming Li.
ABC: A Practicable Sketch Framework for Non-Uniform Multisets. [link]
IEEE International Conference on Big Data (BigData) 2017.

Workshop and Demo

[29] **Yang Zhou**, Hao Jin, Peng Liu, Haowei Zhang, Tong Yang, and Xiaoming Li.
Accurate Per-Flow Measurement with Bloom Sketch. [link]
IEEE International Conference on Computer Communications Workshops (INFOCOM WKSHPS) 2018.

Journal

[30] Zhuochen Fan, Gang Wen, Zhipeng Huang, **Yang Zhou**, Qiaobin Fu, Tong Yang, Alex X Liu, and Bin Cui.
On the Evolutionary of Bloom Filter False Positives - An Information Theoretical Approach to Optimizing Bloom Filter Parameters. [link]
IEEE Transactions on Knowledge & Data Engineering 2022.

[31] Yuanpeng Li, Xiang Yu, Yilong Yang, **Yang Zhou**, Tong Yang, Zhuo Ma, and Shigang Chen.
Pyramid Family: Generic Frameworks for Accurate and Fast Flow Size Measurement. [link]
IEEE/ACM Transactions on Networking 2021.

[32] Tong Yang, Jie Jiang, **Yang Zhou**, Long He, Jinyang Li, Bin Cui, Steve Uhlig, and Xiaoming Li.
Fast and Accurate Stream Processing by Filtering the Cold. [link]
The VLDB Journal 2019.

[33] Tong Yang, Jie Jiang, Peng Liu, Qun Huang, Junzhi Gong, **Yang Zhou**, Rui Miao, Xiaoming Li, and Steve Uhlig.
Adaptive Measurements Using One Elastic Sketch. [link]
IEEE/ACM Transactions on Networking 2019.

[34] **Yang Zhou**, Omid Alipourfard, Minlan Yu, and Tong Yang.
Accelerating Network Measurement in Software. [link]
ACM SIGCOMM Computer Communication Review 2018.

TALKS

- UCCL: Building Efficient GPU Communication Systems
  UC Berkeley SkyLab Winter Retreat — *January 2026*
  UIUC — *Dec 2025*
  Northeastern, UW-Madison — *Nov 2025*
  Google network summit, Google NetInfra — *Oct 2025*
  Meta, ByteDance (Seed), SJTU IPADS — *July 2025*
  ByteDance (Networking), NVIDIA — *June 2025*
  UC Berkeley SkyLab Summer Retreat, Broadcom — *May 2025*
  UC Berkeley SkyLab Winter Retreat — *January 2025*
- Network-Application Co-design for Efficient Datacenters
  University of Toronto — *April 2024*
  NYU, Brown, UC Irvine, UWaterloo, UC Davis, Boston University — *March 2024*
  UC Santa Cruz, University of Virginia, Purdue — *February 2024*
- Electrode: Accelerating Distributed Protocols with eBPF
  Duke University, ACE Center for Evolvable Computing, Google, USENIX NSDI — *April 2023*
  Columbia University — *March 2023*
- Carbink: Fault-Tolerant Far Memory
  Cornell University — *November 2023*
  WORDS workshop — *November 2022*
  Microsoft Research Redmond, USENIX OSDI — *July 2022*
  Google — *March & June 2022*
- Evolvable Network Telemetry at Facebook
  USENIX NSDI — *April 2022*
  Boston University, Meta — *March 2022*
- Cold Filter: A Meta-Framework for Faster and More Accurate Stream Processing
  Harvard University — *October 2018*

STUDENTS

Current Students

- Shuang Ma (PhD) — *2025 - now*
- Yihan Zhang (PhD) — *2025 - now*

Mentored Students

- Ziming Mao, UC Berkeley PhD *2024 - now*
  UCCL (2× OSDI 2026).
- Zhongjie Chen, Tsinghua University PhD → MSRA senior researcher *2024 - now*
  Extensible kernel transport (NSDI 2025), (2× OSDI 2026).
- Xuanlin Jiang, Peking University undergraduate → Harvard PhD *2024*
  CPU offloading for online LLM inference (MLSys 2025).
- Matt Kiley, Harvard College undergraduate → Clockwork Systems *2023*
  Accelerating distributed transactions using eBPF (NSDI 2024).
- Yunxi Shen, Tsinghua University undergraduate → Cornell PhD *2023*
  Resource-efficient job scheduling in data centers.
- Xingyu Xiang, Peking University undergraduate → Harvard PhD *2023*
  Accelerating distributed transactions using eBPF (NSDI 2024).
- Zezhou Wang, Peking University undergraduate → University of Washington PhD *2022*
  Accelerating Paxos using eBPF (NSDI 2023).

Grants

- AWS Research Credits on GPUs, $477,000 *2025*

Teaching Experience

- **Instructor** ECS 150: Operating Systems (with Sam King), UC Davis *Spring 2026*
- **Instructor** ECS 289D: Seminar in Operating Systems: Datacenter Systems for LLMs, UC Davis *Fall 2025*
- **Guest Lecture** on UCCL, CS538: Advanced Computer Networks, UIUC *Dec 2025*
- **Guest Lecture** on UCCL, CS7670: Large Language Model Systems (Seminar), Northeastern *Nov 2025*
- **Guest Lecture** on far memory, CS294-252: Architectures and Systems for Warehouse-Scale Computers, UC Berkeley *Nov 2023*
- **Teaching Assistant** for Prof. Minlan Yu, CS145: Networking at Scale, Harvard University *Spring 2021*
- **Teaching Assistant** for Prof. Tong Yang, Algorithm Design and Analysis, Peking University *Fall 2018*

Patents

- **Yang Zhou**, Hassan Wassel, Minlan Yu, Hank Levy, David Culler, and Amin Vahdat. "Fault Tolerant Disaggregated Memory". Pending (US20230185666A1), December 2022.

Awards and Honors

- Google Ph.D. Fellowship in Systems and Networking *2022*
- Finalist, Meta Ph.D. Fellowship in Networking *2022*
- Graduate Fellowship, Harvard University *2018*
- Excellent Bachelor Thesis (10/327), School of EECS, Peking University *2018*
- New Academic Star Award (1/193), School of EECS, Peking University *2018*
- Arawana Scholarship (2/193), Peking University *2017*
- Pinyou Hudong Scholarship, School of EECS, Peking University *2016*

- May Fourth Scholarship, Peking University *2015*